



FULL-STACK WEB DEVELOPER - JAVA (FSWDJV)

syllabus

FULL-STACK WEB DEVELOPER -JAVA(FSWDJV)

DURATION: 6 MONTHS / 216 HOURS

OBJECTIVE

The Objective of this course is to learn the core concepts of both the frontend and backend programming. This is a valuable Course for Students, programmers and developers which helps them to build dynamic, rich web sites and web applications using compiled language like JAVA.

EXIT PROFILE

- ✓ Complete Knowledge in JAVA
- ✓ Front end development
- ✓ Designing level knowledge

CAREER PATH

- ✓ JAVA Full-Stack Developer
- ✓ Frontend developer
- ✓ UI Designer

FACULTY SKILL SET

- ✓ UI Designing
- ✓ HTML, CSS
- ✓ JavaScript
- ✓ jQuery
- ✓ Strong Knowledge in Core JAVA & Web Frameworks

HARDWARE AND SOFTWARE REQUIREMENTS

- ✓ Operating System: Windows 7/8/10
- ✓ Minimum Memory: 4 GB
- ✓ Recommended Memory: 4 GB / 8 GB
- ✓ Minimum Disk Space: 160 GB
- ✓ Recommended Disk Space: 500 GB
- ✓ Software: Java SDK, Eclipse, Apache web Server, Web Frameworks

PREREQUISITE FOR STUDENTS

Before attending this course, students must have:

- ✓ Basic Knowledge of HTML, CSS
- ✓ Knowledge in web programming will be added advantage

COURSE OUTLINE			
COURSE NAME:	FULL STACK WEB DEVELOPER -JAVA(FSWDJV)	DURATION:216 H/6 M	
MODULE	TOPICS	DURATION	TOTAL
Module - I	Front end development: UI designing	6H/4D	90 H/2.5 M
	HTML	12H/8D	
	CSS	12H/8D	
	JavaScript	30H/20D	
	jQuery	18H/12D	
	Bootstrap	12H/8D	
	Core Java Programming	33H/22D	
Module - II	RDBMS - SQL Server Programming	24H/16D	126H/3.5M
	JDBC	9H/6D	
	Servlets	9H/6D	
	JSP	9H/6D	
	Spring Framework	9H/6D	
	Hibernate	9H/6D	
	Cloud Computing: GitHub, AWS, Azure	6H/4D	
	Working with Servers	6H/4D	
	Project Work	12H/8D	

MODULE IN DETAIL

FRONT-END DEVELOPMENT

UI DESIGNING

- ✓ Colour Concepts
- ✓ Menus
- ✓ Tools
- ✓ Colour Theory
- ✓ Typography
- ✓ Introduction to UI design
- ✓ Real-time designing of a Professional Website.

HTML

- ✓ Introduction To HTML

- ✓ Basic HTML TAGS
- ✓ Advanced HTML Tags & concepts

CSS

- ✓ Introduction to CSS
 - What is CSS?
 - CSS Syntax
 - Three ways to write CSS
 - CSS Priorities
 - Cascading Effect of CSS Properties
- ✓ CSS Basics
 - CSS Selectors
 - Basic CSS Properties (Color, Background, Padding, Margin, Width, Height, Font properties etc)
 - Box Model
 - Units and Colors (px % etc. named colors Hex color representation etc)
 - Using Browser tools to Check CSS Properties Rendered.
- ✓ CSS Advanced
 - Max and min widths, Inline Block, z-index, align etc
 - Pseudo Classes
 - Box Sizing, Display, Position etc
 - CSS Layouts
 - Flex and Grid
 - CSS Transformations
 - CSS Animations
 - Responsive Design (Media Queries)

JAVASCRIPT

- ✓ Introduction to DHTML
 - What is DHTML
 - History of Different ways dynamic elements were added to HTML Pages
 - ✓ JavaScript Basics
 - Writing JavaScript in Browser Console
 - Writing JavaScript in Web pages
 - Linking External
 - Using Console.log to debug the script ins browser Console.
 - ✓ JavaScript Advanced
 - Variables, operators, datatypes, loops, Conditions
 - Objects and Arrays
 - Accessing Arrays and Objects
 - Functions
 - Functions within Objects
 - JSON
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- ✓ Document Object Model
 - What is DOM
 - Viewing the DOM in Browser Developer Tools
- ✓ Manipulating DOM using JS
 - Accessing document object
 - Accessing elements using ID, Class and tag
 - Manipulating existing elements
 - Creating and adding New Elements to document object
- ✓ DOM Events
 - What are events
 - Different types of events
 - Call a function on onclick event
 - Adding event listeners
 - Form validation Using JS

JQUERY

- ✓ jQuery Basics
 - What is jQuery?
 - Adding jQuery to a web page
 - jQuery Syntax
 - jQuery Selector
 - jQuery Events
- ✓ jQuery Advanced
 - HTML Manipulations
 - Effects
 - Traversing
 - Plugins

BOOTSTRAP

- ✓ What is boot strap?
- ✓ Adding bootstrap to your projects.
- ✓ Different predefined Element styles
- ✓ The Grid System
- ✓ Forms
- ✓ Templates

BACK-END DEVELOPMENT USING JAVA

CORE JAVA

- ✓ IDE Tools– Eclipse (Installing & Features)
- ✓ Overview
- ✓ Features of JAVA
- ✓ Programming Structures in Java
- ✓ Arrays
- ✓ Strings

- ✓ Collection Types
- ✓ Wrapper classes
- ✓ Utility Classes
- ✓ OOPS Features
- ✓ Exceptions
- ✓ Multithreading

RDBMS –SQL SERVER

- ✓ SQL RDBMS Concepts
- ✓ SQL Data Types
- ✓ SQL Syntax
- ✓ SQL Operators
- ✓ SQL Joins
- ✓ Views
- ✓ Triggers, Cursors
- ✓ Stored Procedure
- ✓ SQL Date Functions
- ✓ SQL Useful Functions
- ✓ SQL Sub Queries
- ✓ Database Backup & Restore

JDBC

- ✓ JDBC drivers
- ✓ Advantages of JDBC
- ✓ Environment Setup
- ✓ Connectivity with MySQL using JDBC

SERVLETS

- ✓ Introduction to Web servers
- ✓ Introduction to Servlets
- ✓ Environment Setup
- ✓ Servlet API
- ✓ Servlet Request & Response
- ✓ Servlet Collaboration
- ✓ Database Operations using Servlets

JSP (JAVA SERVER PAGES)

- ✓ Introduction Web servers & Servlets
- ✓ Environment Setup
- ✓ Life Cycle
- ✓ Working with servlets
- ✓ Servlet Forms
- ✓ Writing Filters
- ✓ Exceptions
- ✓ Cookies Handling
- ✓ Session Tracking



- ✓ Database Access
- ✓ File Uploading
- ✓ Handling Date
- ✓ Page Redirection
- ✓ Hits Counter
- ✓ Auto Refresh
- ✓ Sending Email
- ✓ Internationalization
- ✓ Annotations

GITHUB

What is Git and Gitlab?

- ✓ History of Git
- ✓ Design Principles
- ✓ Distributed Version Control

Installing Git

- ✓ Account Setup
- ✓ Installing Source tree Git GUI

Git File Management

- ✓ Common Git Commands
- ✓ Configuring Git
- ✓ Creating Repositories
- ✓ Creating a Commit

Branching

- ✓ Visualizing Branches
- ✓ Branch Naming Conventions
- ✓ Creating a new Branch
- ✓ Handling Merge Conflicts

Pull Requests

- ✓ Creating a Merge Request
- ✓ Accepting a Merge Request
- ✓ Rejecting a Merge Request

Common Workflows

- ✓ Centralised Flow
- ✓ GitHub Flow
- ✓ Git Flow

WORKING WITH SERVERS

- ✓ Project Hosting in Internet
- ✓ Payment Gateway Integration
- ✓ API Integration
- ✓ Site lock
- ✓ SSL

PROJECT WORK